WORKED SOLUTION

STEP I 1998, Question 13

Probability and Statistics • Sampling With Replacement

STEP I 1998, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q13

The key idea

This is a conditional probability: the event we condition on is "exactly one pastille eaten in n draws", and we split it up according to *when* that pastille was eaten. Each piece is a product of simple probabilities, because the bag's contents only change at the moment a red is eaten. Adding the pieces is a geometric series, and that's where the difference of two nth powers comes from.

Solution

Write $N = r + b$ for the number of pastilles at the start. We assume $r \geq 1$ and $b \geq 1$ (if $b = 0$ and $n \geq 2$, eating exactly one pastille is impossible and there is nothing to condition on).

The bag at each draw. A non-red pastille goes back, so the bag changes only when a red is eaten. Before the red is eaten the bag holds N pastilles, b of them non-red; afterwards it holds $N - 1$ pastilles, still b of them non-red. Because the bag is shaken thoroughly before each draw, each draw is a fresh uniform choice from whatever is in the bag at that moment, so we can multiply the probabilities draw by draw.

Splitting by the time of eating. For $k = 1, 2, \ldots, n$ let E_k be the event that the only pastille eaten in the n draws was eaten on draw k. Then draws $1, \ldots, k-1$ are non-red from the full bag, draw k is red from the full bag, and draws $k+1, \ldots, n$ are non-red from the reduced bag:

$$\mathrm{P}(E_k) = \left(\frac{b}{N}\right)^{k-1} \frac{r}{N} \left(\frac{b}{N-1}\right)^{n-k} = \frac{r\, b^{n-1}}{N^k\, (N-1)^{n-k}}.$$

The events $E_1, \ldots, E_n$ are disjoint, and their union is the event E that exactly one pastille was eaten. So

$$\mathrm{P}(E) = r\,b^{n-1} \sum_{k=1}^{n} \frac{1}{N^k (N-1)^{n-k}} = \frac{r\,b^{n-1}}{N^n (N-1)^n} \sum_{k=1}^{n} N^{n-k}(N-1)^k,$$

where we multiplied top and bottom of each term by $N^n(N-1)^n$.

Ask yourself How do we add up $\sum_{k=1}^{n} N^{n-k}(N-1)^k$?

It is a geometric series with first term $N^{n-1}(N-1)$ and common ratio $\frac{N-1}{N}$. Even quicker, it is the familiar factorisation

$$x^n - y^n = (x-y)(x^{n-1} + x^{n-2}y + \cdots + y^{n-1})$$

with $x = N$, $y = N-1$, so that $x - y = 1$:

$$\sum_{k=1}^{n} N^{n-k}(N-1)^k = (N-1) \sum_{j=0}^{n-1} N^{n-1-j}(N-1)^j = (N-1)(N^n - (N-1)^n).$$

Hence

$$\mathrm{P}(E) = \frac{r\,b^{n-1}(N^n - (N-1)^n)}{N^n (N-1)^{n-1}}.$$

The conditional probability. The event "I ate it on the last draw, and ate only one" is E_n, with $\mathrm{P}(E_n) = \dfrac{r\,b^{n-1}}{N^n}$. Since $E_n \subseteq E$,

$$\begin{aligned}\mathrm{P}(E_n \mid E) = \frac{\mathrm{P}(E_n)}{\mathrm{P}(E)} &= \frac{r\,b^{n-1}}{N^n} \cdot \frac{N^n (N-1)^{n-1}}{r\,b^{n-1}(N^n - (N-1)^n)} \\ &= \frac{(N-1)^{n-1}}{N^n - (N-1)^n} = \frac{(r+b-1)^{n-1}}{(r+b)^n - (r+b-1)^n},\end{aligned}$$

which is the required result.

Notice that r and b cancel completely, apart from their total. In fact the working shows more: given E, the probability that the pastille was eaten on draw k is proportional to $(\frac{N-1}{N})^k$, so earlier draws are *more* likely than later ones. That makes sense: eating early leaves a bag in which non-reds are a larger fraction, which makes the remaining run of "no more reds" more probable.

Checks

- $n = 1$: the formula gives $\dfrac{1}{N-(N-1)} = 1$, correct, since the only draw is the last one.

- $n = 2$: it gives $\dfrac{N-1}{2N-1}$, which is a little less than $\frac{1}{2}$, consistent with earlier draws being slightly more likely.
- With $r = 3$, $b = 5$, $n = 4$ the formula gives $343/1695 \approx 0.2024$, against $\frac{1}{4}$ if all four draws were equally likely. A simulation of 4×10^5 runs gave 0.2028, and exact enumeration of every draw sequence, tracking the bag, reproduced the formula exactly for all $1 \leq r, b \leq 4$ and $1 \leq n \leq 8$.

Common slips

- Using b/N for the non-red draws *after* the red has been eaten. The bag now has $N-1$ pastilles, and this is the whole point of the question: with b/N throughout every E_k would have the same probability and the answer would be $1/n$.
- Forgetting to condition: $\mathrm{P}(E_n) = rb^{n-1}/N^n$ on its own is not the answer, because we are told that exactly one pastille was eaten.

Beyond STEP

The bag here is a simple Markov chain: its state is the number of reds left, and the chance of moving down depends only on the current state. Conditioning on the end state of such a chain and asking about the path it took is the idea behind "hidden Markov models" and Bayesian inference about histories. Grimmett and Welsh, *Probability: An Introduction*, covers conditional probability and Markov chains at exactly this level, and Blitzstein and Hwang, *Introduction to Probability*, has many problems in the same spirit.

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